Standard-Slope Integration:
A New Approach to Numerical Integration

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Abstract

The definite integral is central to measuring area under a curve, yet explicit antiderivatives are often unavailable. Numerical integration is therefore essential, though many classical formulas are cumbersome or yield low-order approximations. Standard-slope integration (SSI) is a derivative-informed method that uses local and boundary-slope information to produce fast, accurate estimates. SSI performs comparably to classical quadrature while avoiding their structural and algebraic complexities.

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1. Introduction

Since its first introduction by Isaac Newton and G. W. Leibniz in the 17th century, the integral has been a central tool of calculus for determining the area under a curve. In practice, however, closed-form antiderivatives are often difficult to obtain because they may be algebraically complicated, require special functions, or fail to exist in elementary form [1]. Even when an antiderivative is available, its evaluation can be cumbersome and may introduce additional computational overhead.

These challenges underscore the need for numerical integration. Classical quadrature formulas — such as the Newton–Cotes family (including the rectangle rule, trapezoid rule, Simpson's Rule, Simpson's 3/8 Rule, and Boole's Rule) and Gaussian quadrature — approximate integrals through polynomial or orthogonal-series interpolation [2]. Although effective, these methods may require many function evaluations, be sensitive to partition width, or inadequately capture local geometric behavior [3, 4].

We introduce Standard-Slope Integration (SSI), a derivative-informed integration method that uses local slope information to approximate the area under a curve [5]. SSI provides a simple and computationally efficient alternative to both antiderivative-based evaluation and traditional quadrature formulas, achieving high accuracy even when closed-form antiderivatives are unavailable or standard formulas become cumbersome. In this way, SSI can rival classical methods while avoiding several of their practical limitations.

Structure of the paper. Section 2 presents the SSI method, including the procedural steps, the foundational equations, and their derivation. Section 3 provides numerical examples that illustrate the method's behavior and performance. Section 4 discusses the results and their implications. Section 5 offers concluding remarks and outlines avenues for future work.

2. Method

The standard-slope method of numerical integration provides a systematic way to determine the area (A) that lies between a curve and the x-axis over a given base interval $[a, b]$, where a and b are the absolute limits of integration.

The base interval is first divided into one or more partitions (subintervals). The slope of the curve is then determined at each point where the curve intersects a partition boundary, and the standard slope ($m°$) of the curve within each partition is calculated. Next, the area under the curve (AUC) within each partition is determined using its respective standard slope. Finally, the areas of all the individual partitions are added together to obtain the total area under the curve.

According to the standard-slope method, the total area (A) under the curve is given by the following general formula.

$$A = \frac{|\Delta x|}{2}\{f(x_a) + f(x_b) + (\rho - 1)[f(x_b) - f(x_a)]\}\,, \tag{1}$$

where Δx is the width of the base interval, ρ (Greek letter *rho*) is the reference area correction factor (simply referred to as the *correction factor*), and a and b represent the lower and upper limits of integration, or vice versa.

By convention, the area is positive for curves that lie above the x-axis and negative for curves that lie below it. In Step 3, we introduce an additional formula derived from Equation (1) that plays a special role in calculating the area under the curve.

We now examine in detail the steps involved in the standard-slope method of numerical integration.

2.1 Steps in Standard-Slope Integration

The standard-slope method of numerical integration consists of three main steps.

Step 1. Estimate the number of partitions

A reliable estimate of the number of partitions (k) into which the base interval $[a, b]$ is to be divided can be found by solving for k according to the following formula.

$$m_a \left(\frac{1 - m^\circ}{m^\circ}\right)^k = m_b \,, \tag{2a}$$

where m_a and m_b are the slopes of the curve at the limits of integration of the base interval, and m° is the standard slope (defined below).
Ideally, slopes m_a and m_b should be within about one order of magnitude of each other. This consideration generally applies to other integration methods such as Simpson's Rule, which would benefit as well.

Rearranging Equation (2a) and solving for k gives

$$k = \frac{\ln\left({}^{m_b}/_{m_a}\right)}{\ln\left({}^{1 - m^\circ}/_{m^\circ}\right)} \,. \tag{2b}$$

Now, take the absolute value of k and round it up to the nearest whole number greater than zero, such that $k \geq 1$. In some cases, it may perhaps be more convenient to round k up or down to the nearest multiple of 10 or 100, for example, as appropriate.

Special Case: Zero and Infinite Slopes

A few general comments are needed for situations in which either m_a or m_b in Equation (2b) is equal to zero or infinity.

• **Zero slope.**
If the correction factor ρ (see Correction Factor) can be determined, then $k = 1$. Otherwise, it is

necessary to substitute a *temporary* working slope in order to determine the number of partitions. The value of this temporary slope should be about one order of magnitude smaller (or larger, if negative) than the other slope (either m_a or m_b) in the pair, but never greater than 1 or less than −1.

• **Infinite slope.**
When the endpoint slope is infinite, it is necessary to assign a new *finite* replacement slope in order to determine the number of partitions. This will generally entail incurring many decimal places in the independent variable, and it may be necessary to use double precision in the calculations.

The *standard slope* ($m°$) is calculated using the following formula, where m_1 and m_2 are the slopes of the curve at the point where the curve crosses the boundaries of the partition (or base interval), namely (x_1, y_1) and (x_2, y_2), respectively.

$$m° = \frac{m_1}{m_1 + m_2}, \tag{3}$$

where $0 \leq m° \leq 1$.

In determining the standard slope, the following rules apply.

• **If** $m_1 = m_2 = 0$, then $m° = 0.5$.
• **If** $m_1 = \pm\infty$, then $m° = 1$.
• **If** $m_2 = \pm\infty$, then $m° = 0$.

Generally speaking, for the purpose of partitioning the base interval, choosing a standard slope ($m°$) that is between 0.4 and 0.6 gives a reasonably good approximation of the AUC. Values closer to 0.5 typically improve accuracy, but the number of partitions (k) will also increase substantially in most cases.

In more practical terms, if the standard slope of the base interval $[a, b]$ is *less* than 0.5, choose a value for the standard slope ($m°$) in Equation (2b) that is between 0.4 and 0.5 (but not equal to 0.5), depending on the degree of accuracy that is desired. On the other hand, if the standard slope is *greater* than 0.5, choose a value for the standard slope that is between 0.5 and 0.6 (but not equal to 0.5), again depending on the degree of accuracy that is desired.

Note: Rather than calculating the number of partitions (k) using Equation (2b), we can simply assign an arbitrary number of partitions based on our best estimate.

Step 2. Determine the partition width

Divide the number of partitions (k) obtained in Step 1 into the length of the base interval $[a, b]$ to obtain the partition width (Δx).

$$\Delta x = \frac{b-a}{k} \tag{4}$$

We are now ready to calculate the area under the curve.

Step 3. Calculate the area under the curve

The total area (A) under the curve (generally an approximate value) can be calculated using the Fundamental Equation of Standard-Slope Integration, derived from Equation (1).

$$A = \sum_{i=1}^{k} \frac{|\Delta x|}{2} \left\{ f(x_{i-1}) + f(x_i) + \left(\frac{2m^\circ{}_i - 1}{3} \right) [f(x_i) - f(x_{i-1})] \right\}, \tag{5}$$

where Δx is the partition width from Step 2, i is the partition index, k is the total number of partitions from Step 1, and $m^\circ{}_i$ is the standard slope on the ith partition.

Using Equation (5), the area under *any* uniform, smooth curve can be estimated with varying accuracy, depending on the initial choice of standard slope. In many cases, however, the *actual* AUC can be determined directly from Equation (1), which uses the correction factor (ρ). While the correction factor may assume any finite value, it typically lies within $0 \leq \rho \leq 2$ (Figure 2). The section Correction Factor presents the derivation of the correction factor (ρ) for powers, roots, and second-order polynomials.

Note: The number of partitions (k) is *always* equal to 1 when using Equation (1), as well as for square/square-root functions and second-order polynomials.

2.2 Formal Derivations

General Formula.

The area under the curve (AUC) for a concave power function, where $f(x) = x^n$ and $n > 0$, can be estimated using a trapezoid, the area (A) of which is equal to

$$A = |\Delta x| \left[\frac{f(x_a) + f(x_b)}{2} \right],$$

where $|\Delta x| = |b - a| = |a - b|$, and a and b are the lower and upper limits of integration, respectively.

The area of the trapezoid *overestimates* the actual area under the curve because part of the trapezoid extends above the curve. At the points where $f(x_a)$ and $f(x_b)$ intersect the curve, we can form a triangle, called the *reference triangle* (Figure 1), consisting of the portion of the trapezoid lying above the curve and a portion of the trapezoid lying below it, the area of which is called the *reference area* (A_{ref}) and is equal to

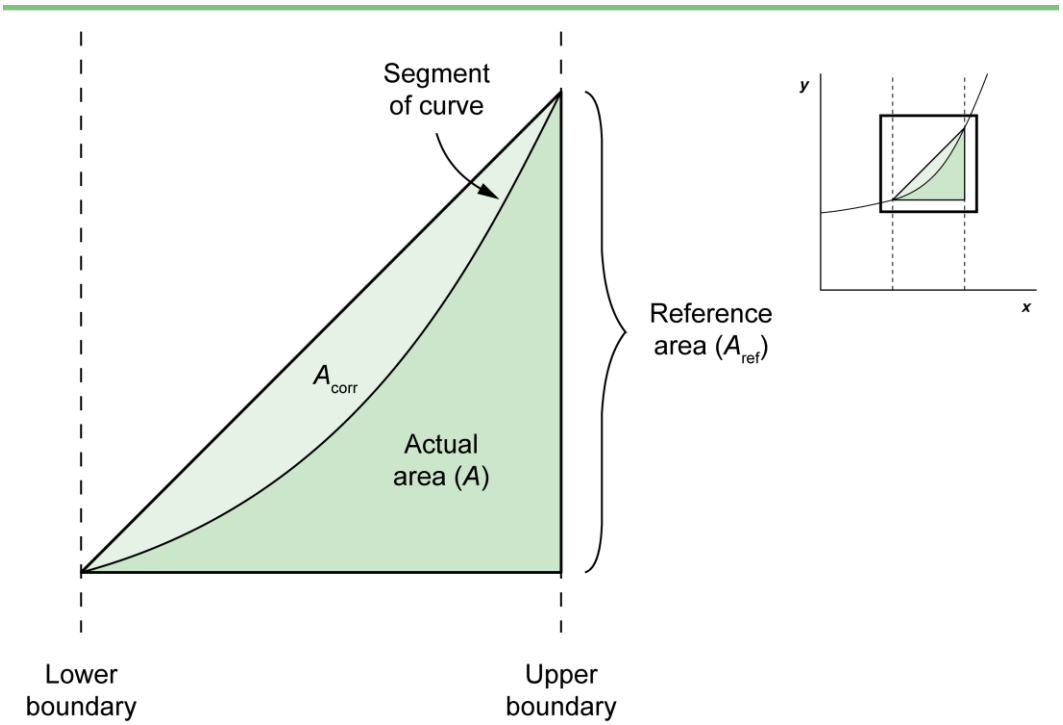

FIGURE 1. Relationship between the reference triangle and its associated areas with respect to a concave curve.
$\boldsymbol{A}_{\mathbf{corr}}$ = correction area

$$A_{\text{ref}} = \frac{1}{2}|\Delta x|[f(x_b) - f(x_a)]\,,$$

where the base of the reference triangle is equal to $|\Delta x| = |b - a| = |a - b|$, and a and b are the lower and upper limits of integration, respectively, that is, the same as for the trapezoid, and its height is equal to $f(x_b) - f(x_a)$.

We use $|\Delta x|$ so that the calculated areas of the trapezoid and reference triangle remain positive regardless of the orientation of the interval or reflection about the y-axis.

Let ρ denote the *reference area correction factor*, where $\rho(A_{\text{ref}})$ is the area of the portion of the reference triangle that lies below the curve, and $(1 - \rho)(A_{\text{ref}})$ is the area of the portion that lies above the curve, which we call the *correction area* (A_{corr}).

The actual area (A) under the curve for a given power function is obtained by subtracting the correction area (A_{corr}) from the area of the trapezoid. (Note: This method can be used to calculate the area under the curve (AUC) for *any* function.)

Therefore,

$$A = |\Delta x|\left[\frac{f(x_a) + f(x_b)}{2}\right] - (1 - \rho)\left\{\frac{1}{2}|\Delta x|[f(x_b) - f(x_a)]\right\}$$

$$= \frac{|\Delta x|}{2}[f(x_a) + f(x_b)] - (1-\rho)\left\{\frac{|\Delta x|}{2}[f(x_b) - f(x_a)]\right\}$$

$$= \frac{|\Delta x|}{2}\{f(x_a) + f(x_b) - (1-\rho)[f(x_b) - f(x_a)]\}$$

$$= \frac{|\Delta x|}{2}\{f(x_a) + f(x_b) + (\rho-1)[f(x_b) - f(x_a)]\}\,.$$

This is precisely the general formula for the standard-slope method introduced in Equation (1).

When using the general formula, the correction area (A_{corr}) is *subtracted* from the area of the trapezoid for a *concave* curve to give the actual area under the curve (AUC). For a *convex* curve, the correction area is *added* to the area of the trapezoid to give the actual area under the curve.

Number of Partitions.

We can get a reasonable estimate of the number of partitions (*k*) into which the base interval should be divided in order to determine the area under the curve (AUC) based on the slope (*m*).

Imagine a straight line, where $y = x$, and the slope (*m*) is the same at any point on the curve. Given the base interval $[a, b]$, slope m_a is equal to slope m_b. In addition, the standard slope ($m°$) is equal to $1/2$, where

$$m° = \frac{m_{\mathrm{a}}}{m_{\mathrm{a}} + m_{\mathrm{b}}}\,.$$

Therefore,

$$m_a\left(\frac{1-m°}{m°}\right)^k = m_b\,,$$

where the term ${}^{1-m°}/_{m°}$ is equal to 1 for a straight line, and *k* is the number of partitions.

This is identical to Equation (2a) in the main text.

As the standard slope ($m°$) approaches $1/2$, $m_a \to m_b$ and the value of *k* approaches infinity for any given curve *except* square/square-root functions and second-order polynomials, where the number of partitions (*k*) is *always* equal to 1.

Standard Slope.

Let us consider the power function $f(x) = x^n$ with $n = 2$. The correction factor ρ (see Correction Factor) for the area under a power function is given by

$$\rho = \frac{2\left(\frac{r^{n+1}-1}{r-1} - n - 1\right)}{(n+1)(r^n - 1)}.$$

For $n = 2$, this becomes

$$\begin{aligned}
\rho &= \frac{2\left(\frac{r^3-1}{r-1} - 3\right)}{3(r^2-1)} \\
&= \frac{2\left(\frac{r^3-1}{r-1} - \frac{3(r-1)}{r-1}\right)}{3(r^2-1)} \\
&= \frac{2\left(\frac{r^3-1-3\mathrm{r}+3}{r-1}\right)}{3(r^2-1)} \\
&= \frac{2(r^3-3\mathrm{r}+2)}{3(r^2-1)(r-1)}.
\end{aligned}$$

Next, we subtract 1 from this correction factor to form the expression $\rho - 1$ that appears in Equation (1).

Therefore,

$$\begin{aligned}
\rho - 1 &= \frac{2(r^3-3\mathrm{r}+2)}{3(r^2-1)(r-1)} - 1 \\
&= \frac{2(r^3-3\mathrm{r}+2)}{3(r^2-1)(r-1)} - \frac{3(r^2-1)(r-1)}{3(r^2-1)(r-1)} \\
&= \frac{2(r^3-3\mathrm{r}+2) - 3(r^3-r^2-\mathrm{r}+1)}{3(r^3-r^2-\mathrm{r}+1)} \\
&= \frac{-r^3+3r^2-3\mathrm{r}+1}{3(r^3-r^2-\mathrm{r}+1)} \\
3(\rho-1)(r^3-r^2-\mathrm{r}+1) &= -r^3+3r^2-3\mathrm{r}+1\,.
\end{aligned}$$

Now, since the relative ratio *r* (see Relative Ratio) of a power function is equal to

$$r = \left(\frac{m_b}{m_a}\right)^{1/(n-1)},$$

and for $n = 2$, we have

$$r = \frac{m_b}{m_a}.$$

Lastly, let m_1 and m_2 denote the slopes at points a and b, respectively.

Next, we substitute m_2/m_1 for the relative ratio (r) in the equation containing the expression $\rho - 1$ and solve for $\rho - 1$.

Therefore,

$$\begin{aligned}
3(\rho-1)\left[\left(\frac{m_2}{m_1}\right)^3 - \left(\frac{m_2}{m_1}\right)^2 - \left(\frac{m_2}{m_1}\right) + 1\right] &= -\left(\frac{m_2}{m_1}\right)^3 + 3\left(\frac{m_2}{m_1}\right)^2 - 3\left(\frac{m_2}{m_1}\right) + 1 \\
3(\rho-1)\left(\frac{m_2^3}{m_1^3} - \frac{m_1 m_2^2}{m_1^3} - \frac{m_1^2 m_2}{m_1^3} + \frac{m_1^3}{m_1^3}\right) &= -\left(\frac{m_2^3}{m_1^3}\right) + 3\left(\frac{m_1 m_2^2}{m_1^3}\right) - 3\left(\frac{m_1^2 m_2}{m_1^3}\right) + \left(\frac{m_1^3}{m_1^3}\right) \\
3(\rho-1)(m_2^3 - m_1 m_2^2 - m_1^2 m_2 + m_1^3) &= -(m_2^3) + 3(m_1 m_2^2) - 3(m_1^2 m_2) + (m_1^3) \\
3(\rho-1)(m_1+m_2)(m_1^2 + m_2^2 - 2m_1 m_2) &= (m_1 - m_2)(m_1^2 + m_2^2 - 2m_1 m_2) \\
3(\rho-1)(m_1+m_2) &= m_1 - m_2 \\
\rho - 1 &= \frac{m_1 - m_2}{3(m_1+m_2)}.
\end{aligned}$$

Finally, we solve the expression for $\rho - 1$ in terms of the standard slope ($m°$), which is given by the formula

$$m° = \frac{m_1}{m_1 + m_2}$$

introduced earlier in Equation (3).

Therefore,

$$\begin{aligned}
\rho - 1 &= \frac{m_1 - m_2 + (m_1 - m_1)}{3(m_1+m_2)} \\
&= \frac{2m_1 - m_1 - m_2}{3(m_1+m_2)} \\
&= \frac{2m_1 - (m_1+m_2)}{3(m_1+m_2)} \\
&= \frac{2m_1}{3(m_1+m_2)} - \frac{m_1+m_2}{3(m_1+m_2)} \\
&= \frac{2m_1}{3(m_1+m_2)} - \frac{1}{3}
\end{aligned}$$

$$= \frac{2m_1/_{m_1+m_2}}{3} - \frac{1}{3}$$

$$= \frac{2\left(m_1/_{m_1+m_2}\right) - 1}{3}$$

$$= \frac{2m^\circ - 1}{3}.$$

Substituting this expression for $\rho - 1$ into the general formula, Equation (1), yields a single-partition form analogous to Equation (5), the Fundamental Equation of Standard-Slope Integration, which sums over all partitions.

It is worth noting that the choice of $n = 2$ is intentional, as higher powers such as $n = 3$ or above produce significantly more cumbersome expressions for $\rho - 1$ without offering additional conceptual insight.

Correction Factor: Powers, Roots, and Second-Order Polynomials.

The area under the curve (AUC) for a given power function, where $f(x) = x^n$, can be found by using the definite integral, where the area (A) is equal to

$$A = \int_a^b x^n \, dx = \left.\frac{x^{n+1}}{n+1}\right|_a^b = \frac{b^{n+1} - a^{n+1}}{n+1}.$$

On the other hand, the general formula in Equation (1) can also be used to find the total area (A) under the curve.

$$A = \frac{|\Delta x|}{2}\{f(x_a) + f(x_b) + (\rho - 1)[f(x_b) - f(x_a)]\},$$

where $\Delta x = b - a$.

If we let $[a, b] = [1, 2]$, then the relative ratio r (see Relative Ratio) is equal to 2, where

$$r = \left[\frac{f(x_b)}{f(x_a)}\right]^{1/n}.$$

Setting the definite integral and the general formula equal to each other and making the appropriate substitutions, we can solve for the correction factor (ρ).

Before doing so, note that the relative ratio (r) is equal to 2; therefore, we can substitute the variable r for the variable b since $b = 2$. In addition, the term $|\Delta x|$ becomes the expression $r - a$, without the absolute value sign.

Therefore,

$$\frac{b^{n+1}-a^{n+1}}{n+1}=\frac{|\Delta x|}{2}\{f(x_a)+f(x_b)+(\rho-1)[f(x_b)-f(x_a)]\}$$

$$\frac{r^{n+1}-1^{n+1}}{n+1}=\left(\frac{r-1}{2}\right)[1^n+r^n+(\rho-1)(r^n-1^n)]$$

$$\frac{r^{n+1}-1}{n+1}=\left(\frac{r-1}{2}\right)[1+r^n+(\rho-1)(r^n-1)]$$

$$\left(\frac{2}{r-1}\right)\left(\frac{r^{n+1}-1}{n+1}\right)=1+r^n+(\rho-1)(r^n-1)$$

$$\left(\frac{2}{r-1}\right)\left(\frac{r^{n+1}-1}{n+1}\right)-1=r^n+(\rho-1)(r^n-1)$$

$$\left(\frac{2}{r-1}\right)\left(\frac{r^{n+1}-1}{n+1}\right)-2=r^n-1+(\rho-1)(r^n-1)$$

$$=[r^n-1][1+(\rho-1)]$$

$$=\rho(r^n-1)$$

$$\left(\frac{2}{r-1}\right)\left(\frac{r^{n+1}-1}{n+1}\right)-\frac{2(n+1)}{n+1}=$$

$$\left(\frac{2}{r-1}\right)(r^{n+1}-1)-2(n+1)=\rho(n+1)(r^n-1)$$

$$\frac{2(r^{n+1}-1)}{r-1}-2(n+1)=$$

$$2\left(\frac{r^{n+1}-1}{r-1}-n-1\right)=$$

$$\rho=\frac{2\left(\frac{r^{n+1}-1}{r-1}-n-1\right)}{(n+1)(r^n-1)}.$$

This equation for the correction factor (ρ) applies to powers and roots of the *n*th order, where $f(x)=x^n$, as well as second-order polynomials (where $n=2$).

Relative Ratio: Powers, Roots, and Second-Order Polynomials.

The correction factor ρ (see Correction Factor) for the power function $f(x)=x^n$ is a function of the order (n) and the relative ratio (r). The expression for the relative ratio (r) can be determined by setting a fixed value for the relative ratio and assigning different values to the order (n).

For illustration, let $n=2,3,4$ and the base interval $[a,b]=[1,2]$. Then slopes $(m_a,m_b)=(2,4),(3,12),(4,32)$, respectively, giving the slope ratios $m_b/m_a=2,4,8$.

Therefore, the relative ratio (r) is given by the formula

$$r = \left(\frac{m_b}{m_a}\right)^{1/(n-1)} ,$$

which in this case is equal to 2.

Now, for powers and roots only, if we proceed as above, then $f(x_a)$ and $f(x_b)$ are equal to $(1, 4), (1, 8), (1, 16)$, respectively, and the ratios $f(x_b)/f(x_a) = 4, 8, 16$.

Therefore, the relative ratio (r) is given by the formula

$$r = \left[\frac{f(x_b)}{f(x_a)}\right]^{1/n} ,$$

which in this case is again equal to 2.

We note several key properties exhibited by the relative ratio, as follows. Slopes m_a and m_b may take on any value that yields the same ratio as m_b/m_a for any given relative ratio (r). The same applies to $f(x_a)$ and $f(x_b)$. Moreover, the relative ratio (r) is the *same* for all orders (n) over the *same* base interval.

3. Numerical Examples

Example 1.

Evaluate

$$\int_1^4 \sqrt{x}\, dx$$

using the standard-slope integration (SSI) method.

The derivative of the base equation is

$$f'(x) = \frac{1}{2} x^{-1/2} .$$

Because the base equation is that of a square root, set the number of partitions (k) equal to 1. Therefore, the width (Δx) of the single partition is

$$\Delta x = \frac{4-1}{1} = \frac{3}{1} = 3 .$$

Next, calculate the area (A) of the partition using Equation (5).

Partition data:

$$f(x_0) = f(1) = 1 \text{ and } f(x_1) = f(4) = 2\,,$$

$$f'(x_0) = f'(1) = {}^{1}\!/_{2} \text{ and } f'(x_1) = f'(4) = {}^{1}\!/_{4}\,,$$

and

$$m^\circ = \frac{{}^{1}\!/_{2}}{{}^{1}\!/_{2} + {}^{1}\!/_{4}} = \frac{{}^{1}\!/_{2}}{{}^{3}\!/_{4}} = \frac{2}{3}\,.$$

Therefore,

$$\begin{aligned}
A &= \frac{|3|}{2}\left\{1 + 2 + \left[\frac{2({}^{2}\!/_{3}) - 1}{3}\right](2-1)\right\} \\
&= \frac{3}{2}\left[3 + \left(\frac{{}^{1}\!/_{3}}{3}\right)(1)\right] \\
&= \frac{3}{2}\left[3 + \left(\frac{1}{9}\right)(1)\right] \\
&= \frac{3}{2}\left(\frac{27}{9} + \frac{1}{9}\right) \\
&= \frac{3}{2}\left(\frac{28}{9}\right) \\
&= \frac{14}{3} \\
&= 4\frac{2}{3}\,.
\end{aligned}$$

The actual value of the integral is $4\ 2/3$.

Comparison with Simpson's Rule.
For this integral, Simpson's Rule requires 18 partitions to achieve an error below 10^{-5}. The resulting approximation is 4.66667, agreeing with the actual value to five decimal places. In contrast, SSI with a single partition already yields the actual value.

Example 2.

Using a standard slope (m°) of 0.46, evaluate

$$\int_0^1 x^2 e^x \, dx$$

to five decimal places.

The derivative of the base equation is

$$f'(x) = 2xe^x + x^2 e^x \, .$$

The base equation is a mixed function. Because slope m_a is equal to zero, we temporarily assign it a different value — say, 0.8 — in order to partition the base interval. Therefore, the number of partitions (k) is

$$m_a = f'(0) = 0 \leftrightarrow 0.8 \text{ and } m_b = f'(1) = 3e$$

and

$$k = \left| \frac{\ln\left({}^{3e}/_{0.8}\right)}{\ln\left({}^{0.54}/_{0.46}\right)} \right| = 14.48 \rightarrow k = 20 \, .$$

Partition width (Δx):

$$\Delta x = \frac{1-0}{20} = \frac{1}{20} = 0.05 \, .$$

Next, calculate the area (A_i) of each partition (p_i) in the usual manner using Equation (5).

Note: Only the area of the first partition is calculated here.

• p_1

Partition data:

$$f(x_0) = f(0) = 0 \text{ and } f(x_1) = f(0.05) = 0.0026282 \, ,$$

$$f'(x_0) = f'(0) = 0 \text{ and } f'(x_1) = f'(0.05) = 0.1077553 \, ,$$

and

$$m^\circ{}_1 = \frac{0}{0 + 0.1077553} = \frac{0}{0.1077553} = 0 \, .$$

Therefore,

$$
\begin{aligned}
A_1 &= \frac{\left|{}^{1}\!/_{20}\right|}{2}\left\{0 + 0.0026282 + \left[\frac{2(0)-1}{3}\right](0.0026282 - 0)\right\} \\
&= \frac{1}{40}\left[0.0026282 + \left(\frac{-1}{3}\right)(0.0026282)\right] \\
&= \frac{1}{40}(0.0026282 - 0.0008761) \\
&= \frac{1}{40}(0.0017521) \\
&= 0.0000438\,.
\end{aligned}
$$

• Total Area (A)

$$A = 0.7182847 \approx 0.71828$$

The actual value of the integral is $e - 2 \approx 0.71828$.

Comparison with Simpson's Rule.
For this integral, Simpson's Rule requires 16 partitions to achieve an error below 10^{-5}. The resulting approximation is 0.71828, agreeing with the actual value to five decimal places. In contrast, SSI requires 20 partitions to obtain the same approximate value.

4. Discussion

Every base equation or function brings with it a distinct integral, and no single technique uniformly excels across all possible forms. Both classical integration and standard-slope integration (SSI) are fundamentally tied to the structure of the underlying function; the essential difference is that classical integration relies on the *antiderivative*, while SSI relies on *derivative* information.

Despite this conceptual departure, SSI exhibits several advantages over established numerical-integration methods.

• **Conceptual and computational simplicity.**
The procedure requires only function values and slopes at partition boundaries, avoiding the algebraic complexity associated with antiderivatives or higher-order interpolation schemes.

• **Accuracy and reliability.**
The method consistently produces results comparable to classical quadrature and competitive with widely used Newton–Cotes formulas.

• **Predictable and controllable error behavior.**
The accuracy can be tuned directly through partition refinement, with the standard slope providing a reliable estimate of the number of partitions required.

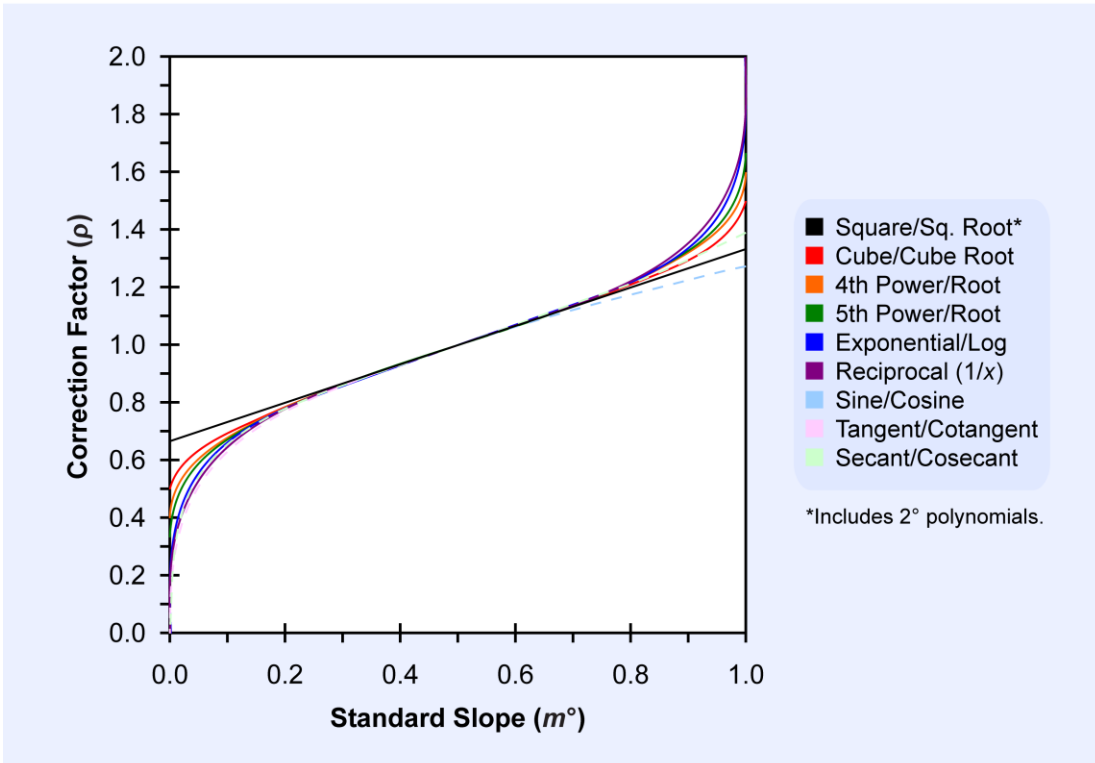

FIGURE 2. Reference area correction curves.

• **Broad applicability.**
SSI applies directly to all differentiable functions and does not rely on special structural assumptions for good performance.

• **Customizable correction curves.**
The framework accommodates tailored correction curves for any differentiable function, enabling users to refine the correction factor whenever additional structural information is available.

The cornerstone of the standard-slope method is the standard slope ($m°$) itself. The underlying principle behind this method is most easily understood through the following observation. Given the base interval $[a, b]$, where a and b represent the limits of integration, there exists a unique uniform, smooth curve that passes through the points (x_a, y_a) and (x_b, y_b) and has slopes m_a and m_b, respectively. Consequently, the slope pair (m_a, m_b) — and in particular the standard slope — determines a unique relationship between the endpoint slopes and the area under the curve for any given base interval.

Furthermore, there exists a direct relationship between the standard slope ($m°$) of the curve and the reference area (A_{ref}) of a circumscribed reference triangle (Figure 1). This relationship is expressed by the reference area correction factor (ρ). From this, a reference area correction curve can be constructed for *any* given uniform, smooth curve (Figure 2). Conversely, if the standard slope ($m°$) and the reference area (A_{ref}) are known, the area under the curve of the associated reference triangle can be determined.

The standard-slope method makes use of this relationship to determine the actual area under the curve. It does so by first calculating the area of the trapezoid over the base interval and then adjusting this value based on the corrected (actual) area under the curve of the associated reference triangle, located at the tip, in order to obtain the actual area (A) under the curve.

To further understand the role of the standard slope ($m°$), it is useful to examine the correction curves for several representative classes of functions shown in Figure 2. These curves reveal a number of structural features that are particularly useful for understanding integration using the standard-slope method.

• The correction curve for square and square-root functions (as well as second-order polynomials) is a straight line.
• The correction curves for other function classes are typically S-shaped.
• The correction curves closely approximate one another for standard slopes ($m°$) between 0.4 and 0.6.
• All correction curves intersect at the same point, namely at a standard slope ($m°$) of 0.5.
• For functions of the form x^n or $x^{1/n}$, the difference between their correction curves decreases as n increases.
• The endpoints of the correction curve for x^n or $x^{1/n}$ can be computed explicitly.

Observe in Figure 2 that the correction curves for a function and its inverse *coincide*, and that the correction curve for any given exponential or logarithmic function is the *same* regardless of the base.

Finally, as noted earlier, the standard-slope method applies to any uniform, smooth curve. A curve ceases to be uniform, however, whenever it changes direction — for example, when it transitions from convex to concave — or when its slope changes sign. Such changes occur at points where $f'(x)$, $f''(x)$, or higher-order derivatives are equal to zero or infinity.

These points constitute what we refer to as *relative limits of integration* (RLIs). Whenever an RLI is present, the area under (over) each resulting segment must be integrated separately, treating the RLI as the *new* limit of integration. In addition to derivative-based RLIs, any point where $f(x)$ itself is equal to zero or infinity also serves as a relative limit of integration.

It is worth noting that, when using the standard-slope method, we can generally integrate in either direction, starting with whichever limit of integration is more convenient. Thus, the base interval may take the form $[a, b]$ or $[b, a]$, depending on the direction of integration, where a and b denote the lower and upper limits, respectively.

It is also important to recognize that integration may involve numbers that are very large or very small, a fact that must be accounted for to ensure the accuracy of the results. In addition, SSI's reliance on local slope information provides practical advantages in settings where derivatives are available or inexpensive to compute. Yet SSI is not derivative-dependent as the slope can be obtained from $\Delta y/\Delta x$ when derivatives are unavailable.

5. Conclusion

The integral remains a foundational tool across physics, engineering, economics, statistics, and many other domains. Yet closed-form antiderivatives are often unavailable, difficult to evaluate, or impractical for routine computation, motivating the development of numerical-integration methods that balance accuracy, efficiency, and general applicability. Standard-Slope Integration (SSI) addresses this need by providing a derivative-informed alternative to classical quadrature rules, employing local slope information rather than antiderivatives or high-order interpolation to approximate the area under a curve.

The method is straightforward to implement, computationally efficient, and yields numerical results comparable to established quadrature rules while avoiding many of their algebraic and structural constraints. These properties position SSI as a practical and broadly applicable addition to the numerical-integration toolkit, particularly in settings where classical formulas become cumbersome or closed-form antiderivatives are inaccessible.

Future Work.
Future work includes applying SSI in settings where derivative information is readily available, such as real-time control, embedded computation, biomedical signal processing, and aerospace guidance. Such studies may clarify the method's practical advantages relative to classical quadrature rules. Additional extensions — such as adaptive step-size strategies (which SSI naturally supports), multidimensional variants, and hybrid integration schemes — offer further opportunities to broaden SSI's utility across diverse computational environments.

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